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Rigless Geothermal Reservoir Evaluation and Abandonment Using Intelligent CT and Advanced BHA Technologies

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Abstract

In the context of geothermal exploration in regions lacking subsurface data, a pioneering rigless methodology was implemented to enable sustainable, data-driven reservoir evaluation. By leveraging intelligent coiled tubing (CT) and real-time-enabled downhole technologies, the intervention aimed to replicate production conditions as closely as possible, allowing accurate assessment of geothermal potential while minimizing operational footprint and environmental impact.

The intervention commenced with RT temperature logging and GR/CCL correlation to TD using intelligent CT. Reverse circulation followed, with nitrogen injected into the annulus while formation well fluids were recovered through the CT, allowing precise volume control and underbalance calibration. A single-run TCP deployment exceeding 100meters was executed using a pressure-activated firing head. Real-time telemetry monitored wellbore parameters and confirmed gun activation, enabling optimized timing and accurate reservoir monitoring post-perforation.

A custom 8.5" OD BHA was deployed, engineered to replicate the performance of a future geothermal production pump. The assembly featured a jet pump based on concentric CT principles and was hydraulically powered via water fluid circulation through CT. A modified compression packer anchored and isolated the pump and included a check valve to isolate the formation upon flow cessation eliminating any wellbore storage effects. Real-time feedback from the embedded tension/compression module ensured accurate packer setting. Below the packer, memory gauges recorded transient formation pressure without hydrostatic interference. The assembly ended with a memory-based water sampling tool, pre-configured with a timing mechanism.

This was followed by deployment of a PLT to acquire a real-time injection profile.

For in-situ formation stress evaluation, a custom-designed straddle packer with reinforced elastomers was used for a multi-layer mini- frac rock strength test. Memory gauges between the packer elements recorded high-resolution pressure data.

A cement retainer with real-time telemetry was then set to perform an extended LOT across the formation cap rock interval and served as a base for fine-cement slurry placement. The solids-free slurry was successfully pumped through the CT, despite the cable, enabling P&A in line with project requirements.

This operational sequence was replicated across all identified zones.

The integrated deployment of intelligent CT with advanced engineered BHAs enabled precise control, monitoring, and comprehensive data acquisition across all key aspects of reservoir behavior—temperature, pressure, fluid dynamics, and formation integrity. Executing complex, multistage operations in a single run improved efficiency and significantly reduced time-on-well.

The successful completion of the full workflow without a rig demonstrated the technical viability of high-resolution geothermal reservoir evaluations.

This intervention represents the first rigless geothermal reservoir assessment using intelligent CT integrated with a suite of bottom-hole technologies. The innovation lies not only in the hardware but in the orchestration of components into a seamless, data-driven, and environmentally responsible workflow.

Introduction

The SCAN project is a government-led initiative funded by the Ministry of Economic Affairs and Climate and executed by EBN, with the strategic objective of acquiring high-quality subsurface data in areas where existing information is insufficient for a reliable estimation of geothermal potential. These target areas have historically seen little or no drilling activity due to their limited oil and gas prospectivity, leaving their geothermal potential largely unexplored. The initiative is a cornerstone of the Netherlands' energy transition strategy, aimed at enabling sustainable, low-carbon heat supply to meet growing demand in metropolitan and industrial areas. The program targets both shallow and deep geothermal resources, excluding ultra-deep plays, at depths ranging from 500 to 4,000 m, with the objective of delivering a robust geological and reservoir dataset to de-risk future developments and promote equitable access to subsurface heat resources across the country.

The project is structured in two main phases:

- I. The first phase focused on subsurface imaging through the acquisition of over 1,700 km of new 2D seismic data, completed between September 2019 and Q1 2022, and the reprocessing of more than 7,500 km of vintage 2D seismic lines, nearly 500 profiles, completed in Q3 2023. This phase delivered a significantly improved geological framework, enhancing the ability to identify prospective intervals and optimize well location selection.
- II. The second phase consists of a drilling campaign involving dedicated data-acquisition wells, with no production planned, to be permanently abandoned after testing. Each well is designed to collect both qualitative and quantitative data to assess reservoir properties, caprock integrity, and in-situ stress conditions, with target intervals selected using a Play-Based Exploration approach to ensure that each location contributes to a dataset with wide spatial influence and coverage of high heat-demand areas.

The well drilled to investigate multiple geothermal plays: the primary target being the Rotliegend Slochteren Formation (ROSL) sandstones, with secondary targets subsurface objectives to prove the presence and determine the reservoir properties of (from top to bottom) the Chalk Group (CK), the Vlieland Sandstone Fm (KNNS) and the Main Buntsandstein Subgp (RBM). Each target includes evaluation of the overlying caprock, and additional data acquired from the formations between and above these intervals. The operational sequence for each SCAN well includes drilling, well testing, extended leak-off tests (X-LOTs) and decommissioning, with the primary goal of delivering a comprehensive subsurface dataset that includes temperature, pressure, permeability, and horizontal stress analysis, before returning the site to its original condition in compliance with local regulations.

The original Scope of Work envisioned the use of coiled tubing only for limited well intervention tasks, such as tubing-conveyed perforating and gas lifting, all in the presence of the rig, while the remainder of well testing and plug and abandonment operations would be conducted using the rig. However, as the campaign

progressed, a combination of cost-control initiatives, sustainability objectives, increased confidence in both the target formations and available technologies, together with advancements in engineering design, prompted a strategic shift in the operational plan. Relying heavily on rig time for post-drilling activities risked eroding the fixed budget, potentially reducing the number of wells and thereby limiting the geographic distribution of subsurface data.

To preserve the scope of the campaign and improve operational efficiency, the project team expanded the role of coiled tubing operations to perform the majority of the post drilling and completion well activities without rig support. By leveraging intelligent CT systems integrated with advanced bottom-hole assemblies and real-time telemetry, the revised workflow enabled the execution of well testing, X-LOTs, and decommissioning entirely in rigless mode. This approach delivered substantial cost savings, minimized surface footprint, and safeguarded the planned well count within the allocated budget, ultimately maximizing both the spatial coverage and the quality of the national geothermal dataset.

Project Background and Well Selection

The well selected for the rigless application, the first in the SCAN campaign to be executed without a rig following drilling and completion, was a relatively shallow data-acquisition well, with a total depth (TD) of 2,093.9 m MD. The completion comprised a 13 5/8" surface casing, transitioning to a 9 5/8" intermediate casing and a 7" liner hanger. All casing transitions were constructed with a step profile, as no entry guides were intentionally installed to keep the design simple for a well that would be permanently abandoned after data acquisition and testing, as shown in the below figure 1.

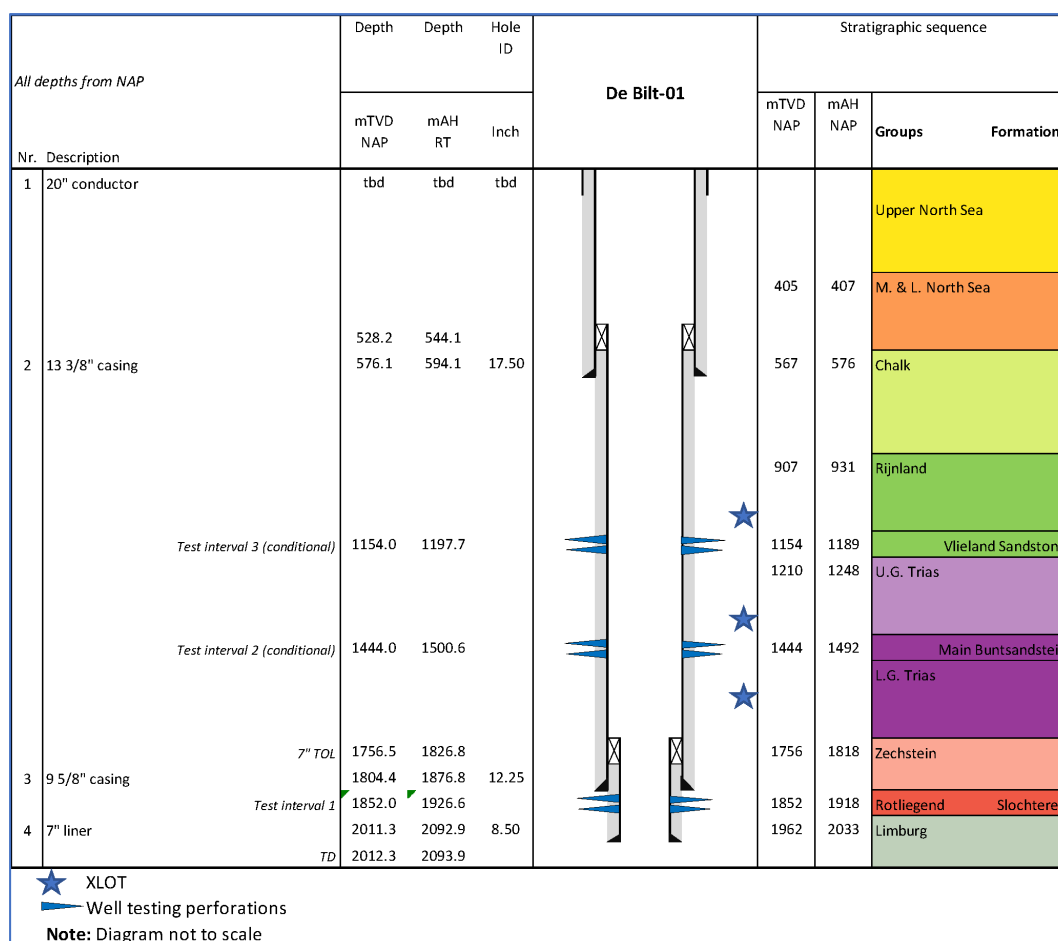


Figure 1—Well Completion Diagram.

The zones of operational interest were located at 2,092 m MD, 1,500 m MD, and 1197 m MD, as shown above. A key objective was to evaluate caprock integrity through horizontal stress measurements, requiring dedicated perforations in specific intervals to perform X- LOT, ensuring controlled fracture initiation and accurate leak-off observation, zone identified in the blue stars in the completion diagram.

The well was positioned on the Zeist Horst, part of the Zandvoort Ridge, a long-lived structural high underlying urban, agricultural, and industrial centers. Despite the high regional heat demand, no wells had previously been drilled in this structural domain, and uncertainty remained high regarding the presence and quality of reservoirs suitable for geothermal exploitation.

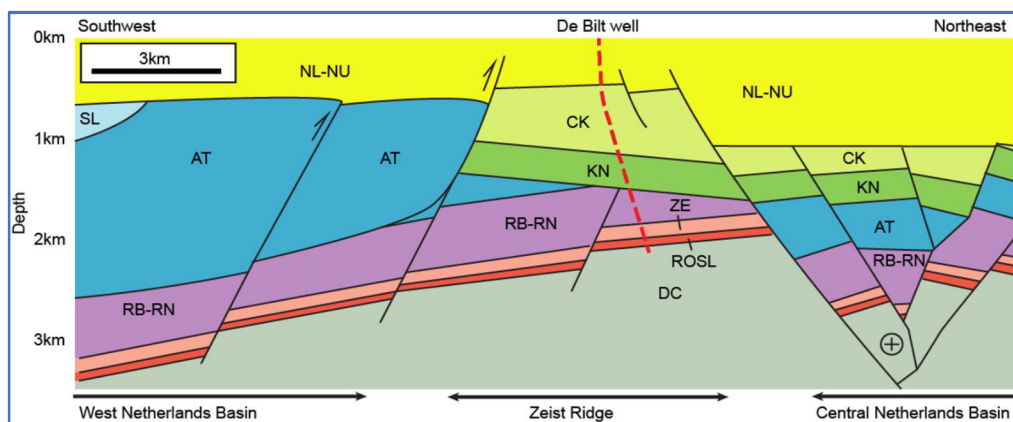


Figure 2—Interpreted composite seismic section showing the anticipated stratigraphy.

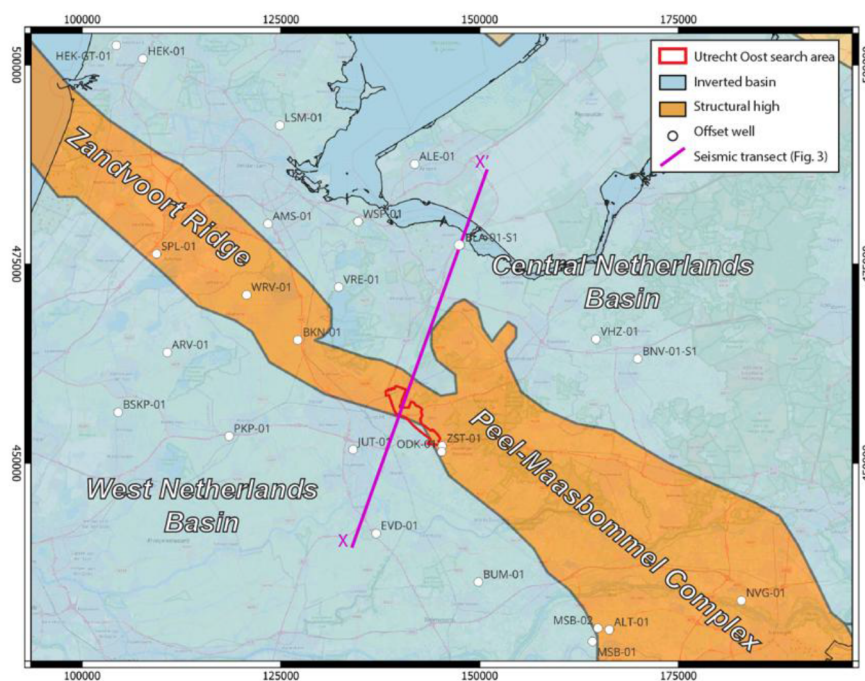


Figure 3—Map showing simplified structural domains around the search area. Location of transect (SCAN002) and relevant offset wells are indicated

The primary target was the Slochteren Formation (ROSL), with secondary objectives including the Chalk Group (CK), the Vlieland Sandstone Formation (KNNS), and the Main Buntsandstein Subgroup (RBM). Each target included evaluation of the overlying caprock, and potential flow zones below the fresh-saline transition were expected to contain brine. Seismic interpretation indicated hydraulic connectivity between certain units, such as the Vlieland Sandstone and the underlying Buntsandstein Subgroup, due

to depositional geometry, and between the Zechstein, Rotliegend, and Limburg Groups owing to vertical stacking and faulting in the area.

The transition from fresh to saline water was expected at ~200 m TVDNAP, with salinity gradually increasing at depth. Formation water salinity in the Vlieland Sandstone and deeper intervals was estimated at 40 g/L Na⁺ and 80 g/L Cl⁻ (TDS = 135 g/L), based on offset wells Jutphaas-01, Blaricum-01-S1, and Everdingen-01. This relatively low salinity likely reflects the absence of Zechstein halite in the area.

Pore pressure analysis from offset wells indicated hydrostatic conditions throughout the planned well profile, with gradients ranging from 1.00 SG EMW in the freshwater aquifer to 1.05-1.08 SG in the deeper formations. No overpressure was anticipated. Minimum horizontal stress (S_{hmin}) was assumed equal to fracture gradient, with estimates based on offset LOT/FIT data and provided in the Basis of Design.

Temperature modelling for the De Bilt area, derived from the latest TNO subsurface temperature model (Veldkamp & Hegen, 2020), predicted 68 °C at the top of the Slochteren reservoir (1,843 m TVDNAP; range: 52-90 °C) and 74 °C at TD (2,001 m TVDNAP; range: 60-95 °C). Despite calibration with offset well data, significant uncertainty remained in the geothermal gradient, making temperature logging a critical element of the data acquisition program.

Building on the geological context, well design, and testing objectives outlined above, the engineering team developed a tailored operational strategy to deliver the required data in a fully rigless mode. This approach demanded careful integration of coiled tubing technology, specialized downhole tools, and real-time data acquisition systems to execute well testing, X-LOTs, and decommissioning activities in a single, efficient sequence. The following section details the engineering considerations, technology selection, and workflow design that underpinned the execution of this first rigless geothermal data well in the SCAN program.

Engineering Desing Phase

The engineering design phase represented the critical first step in enabling a fully rigless execution of the well testing, X-LOT, and decommissioning activities. The challenge was to evaluate the operational limits and capabilities of coiled tubing technology, and to determine which data could be reliably acquired without rig support, while ensuring that the solution remained technically feasible, operationally safe, and economically sustainable. A failure to acquire the full dataset during the rigless phase would have required (re)mobilization of the drilling rig, resulting in significant delays, additional costs, and potential loss of operational momentum.

Engineering efforts focused on identifying and mitigating the constraints imposed by the well's existing configuration, including completion design, downhole geometry, and formations challenges. A detailed feasibility analysis was conducted to assess the ability of coiled tubing to deploy and operate custom-engineered bottom-hole assemblies (BHAs) required to perform all planned operations, including production simulation, reservoir characterization, caprock stress testing, and final abandonment.

The design process prioritized a sustainable and fit-for-purpose approach, avoiding unnecessary complexity while ensuring that all functional requirements could be met. Each operational stage was modeled to anticipate downhole forces, pressure regimes, and fluid dynamics, ensuring that the CT string could deliver the necessary mechanical conveyance, hydraulic power, and data acquisition capability within safe operating envelopes.

The starting point for this workflow was the adoption of an Hybrid Coiled Tubing Technology (HCTT) platform, a hybrid configuration capable of combining the mechanical flexibility of conventional CT with continuous, real-time downhole data transmission. This enabled a dynamic, data-driven approach, allowing operational parameters to be monitored, validated, and adjusted on the fly, ensuring optimal execution of all three project phases under the specific well conditions.

HCTT: Overview of Enabling Technology

To enable the rigless execution the project deployed an HCTT. This technology integrates mechanical conveyance, real-time telemetry, and downhole power delivery in a single intervention platform, combining the flexibility of coiled tubing with the diagnostic and control capabilities traditionally associated with electric-line or fiber-optic wireline systems.

At the core of the HCTT system is a 4.0 mm OD hybrid coaxial cable, embedded in a free-floating configuration along the full length of the coiled tubing string. The cable armor is manufactured from corrosion-resistant Alloy 825, ensuring high tensile strength, superior fatigue resistance, and full compatibility with several fluids, and the bending radius imposed by the CT. This conductor assembly is comprised of:

- Electrical Path - A 17 AWG copper conductor, delivering continuous DC power to downhole tools via power-line communication protocols. This allows sustained operation of high-consumption tools without battery limitations, enabling extended-duration interventions without degradation of performance or loss of data integrity.
- Optical Path - A Fiber-in-Metal Tube (FIMT), constructed from SS316 or Alloy 825 for mechanical robustness and corrosion resistance, housing two single-mode (SM) and two multi-mode (MM) optical fibers. The fibers are hermetically coated for high-temperature (HT) operation, with optical attenuation ≤ 0.4 dB/km at 1310 nm and ≤ 0.26 dB/km at 1550 nm for SM, and ≤ 3.0 dB/km at 850 nm and ≤ 1.1 dB/km at 1300 nm for MM. These specifications ensure high-fidelity optical transmission for Distributed Acoustic Sensing (DAS) and Distributed Temperature Sensing (DTS) in geothermal applications, as well as high-bandwidth data transfer for future advanced diagnostics.

To complete the technology and fully exploit the capabilities of the hybrid cable, the real-time sensing Bottom Hole Assembly (BHA) was deployed. This modular assembly integrates multiple high-precision measurements and correlation subsystems designed to function reliably in the large-diameter and often complex geometries typical of geothermal completions.

Among the most critical components is the Active Casing Collar Locator (ACCL) – a non-magnetic, dual-coil system capable of accurately detecting casing collars at low conveyance speeds and in large-diameter completions where conventional magnetic CCLs frequently fail. The ACCL generates a controlled electromagnetic field; variations in casing geometry or metallurgy cause changes in inductive coupling, producing sharp and repeatable correlation spikes. The dual-coil design allows log-up/log-down processing, which suppresses transient noise and enhances depth correlation accuracy to sub-decimeter resolution.

The sensing BHA also includes:

- Pressure Sensors - Dual-path, internal and annular pressure transducers rated to 15,000 psi, with a resolution of $\pm 0.02\%$ F.S., enabling high-fidelity AP measurements during injection, production, and packer inflation sequences.
- Temperature Sensors - High-precision (± 1 °C) RTD-based elements capable of tracking thermal transients during both fluid displacement and static phases, allowing real-time thermal front mapping.
- Tension / Compression Module - A pressure-balanced load cell with 20 lbs resolution over a 50,000 lbs range, immune to internal CT pressure effects, providing accurate Weight-on-Bit (WOB) measurements and enabling real-time slack-off detection during run-in-hole.
- Dynamic Sensors - Triaxial accelerometers and gyroscopes for vibration, shock, and inclination monitoring, combined with torsional sensors for torque-on-bit detection. All dynamic data are synchronized via the surface telemetry clock, providing a detailed map of downhole toolstring behavior.

The telemetry backbone employs a bi-directional, high-speed digital link between the surface acquisition unit and the downhole toolstring, enabling continuous real-time data flow from all sensors while maintaining full downhole control and actuation capabilities. This architecture supports faster decision-making and allows on-the-fly adjustments of key operational parameters, such as pumping rates, tool positioning, and tension or compression readings, without the need to retrieve the CT string. The surface acquisition system incorporates advanced visualization and analytics, merging log-up/log-down CCL overlays with real-time AP and AT trending, WOB tracking, and event markers for critical operations such as perforation, fracture initiation, and flow transitions.

By combining full mechanical conveyance, high-rate fluid pumping, nitrogen lifting, and jetting with the ability to acquire high-accuracy reservoir and wellbore parameters in real time, the ICT/HCTT platform provides both intervention and diagnostic capabilities within a single deployment. Continuous high-power delivery to electrically driven downhole tools, coupled with high-speed, low-latency data transmission, ensures adaptive control and operational optimization throughout the intervention. This integration eliminates the need for separate wireline operations, reducing surface footprint, rig-up/rig-down time, and overall costs. For the SCAN project, it enabled the complete execution of the three main operational stages, well testing, X-LOT, and decommissioning, in a single, continuous, entirely rigless sequence, without compromising data quality, operational control, or safety margins.

Well Testing Engineering Phase

The well testing phase presented a high level of operational complexity, driven by the requirement to perform a well testing phase, followed by a direct analysis of the reservoir performance, ideally replicating as best as possible the real well production phase. Beyond simple flow testing, the objectives included determining the contribution of each extended pay zone, quantifying the relative injectivity of individual intervals, and confirming the downhole temperature profile. Temperature data acquisition was important, as this is a geothermal well and accurate measurement was essential both to validate the geological interpretation and to establish the geothermal gradient.

One of the most critical operational challenges in the well testing phase, also highlighted by the results from the first well in the SCAN program, was the need to produce the well using artificial lift and then acquire undisturbed reservoir data by shutting down production within seconds. In the previous well, the inability of the reservoir to sustain the hydrostatic head meant that, immediately after stopping the artificial lift, the reservoir starts taking the well bore fluid, cooling down the reservoir, causing partial or total re-entry of column fluids into the reservoir. This effectively partially invalidated post-shutdown reservoir pressure measurements and significantly reduced data quality.

The first engineering challenge was to develop a system capable of maintaining artificial lift while isolating the reservoir almost instantaneously, within few seconds after stopping the lift. Achieving this was critical to prevent backflow of wellbore fluids into the reservoir, which would compromise post-shutdown pressure transient data. Several conceptual solutions were evaluated during the design phase.

One of the initial options relied on a relatively low-complexity approach: a nitrogen-lift system integrated with an invertible mechanical- set packer (Model B), positioned so that the annulus could be isolated during lifting operations simply by moving the coiled tubing either in compression or in tension, depending on the desired packer configuration. Nitrogen would be injected through a dedicated circulation port located above the packer, enabling continuous lift while the reservoir remained hydraulically isolated from the wellbore above. Once isolation was confirmed, nitrogen injections could be stopped, and reservoir response would be recorded by memory gauges placed below the packer. While technically feasible, this configuration was subject to some operational constraints. The mechanical packer's dimensional limitations meant that testing multiple intervals would require either 7" completions or custom-manufactured large-bore packers for the 9" casing sections, with uncertainties on its performance when run with the CT. Moreover, reliance

on nitrogen lift introduced an artificial flow regime unrepresentative of standard geothermal production systems, reducing the relevance of the acquired performance data for future field development scenarios.

A second alternative explored the deployment of electrically actuated inflatable packers, triggered via an electrical impulse through the HCTT conductor system. The concept mirrored the operational sequence of the mechanical packer approach: nitrogen lift would be maintained during production, and upon receiving the activation signal, the inflatable packer would expand to isolate the reservoir. Once isolation was achieved, nitrogen injection would cease, and memory gauges positioned below the packer would record the pressure transient response. Although technically attractive and feasible, offering precise activation control and compatibility with the existing HCTT infrastructure, this solution presented notable drawbacks. The system's complexity and higher capital cost reduced its sustainability for a multi-well campaign, especially for geothermal exploration well with (in this case) no direct ROI. Operational reliance on nitrogen lift remained an issue, as it did not replicate typical geothermal artificial lift conditions. Furthermore, the inherent limitations of inflatable elements, including potential sealing imperfections in proximity of perforated cased hole which could have been damaged the rubbers elements, added an additional layer of operational risk.

While the other two solutions were retained as contingency options to safeguard and confirm the technical feasibility of the rigless phase with the CT, the solution ultimately selected, considered the most reliable in terms of production performance and reservoir isolation efficiency, was the deployment of a custom-designed downhole jet pump assembly, powered by water and engineered to isolate the formation almost instantaneously after ceasing the water injection. The jet pump design was derived from proven concentric coiled tubing BHA configurations, adapted for large-bore geothermal applications, and required only water for activation. This eliminated the cost and logistical burden of nitrogen supply while delivering an artificial lift mechanism representative of actual geothermal production systems. By replicating field production conditions and enabling near-immediate formation isolation, this approach mitigated the risk of data distortion during post-shutdown reservoir monitoring.

Given that the system's operational performance would be dictated by achievable flow rates through the CT, a detailed hydraulic simulations were carried out using two independent simulation platforms to cross-verify results. The analysis encompassed low-case, mid-case, and high-case production forecasts (see [tab 1](#), [2](#) and [3](#)), benchmarked against expected well deliverability, to ensure the selected design could meet or exceed the anticipated range. The worst-case scenario determined the lower operational boundary for pump performance and was instrumental in selecting the coiled tubing size.

Table 1—Production test Slochteren - Low Case

Step	Description	Injection pressure [bar]	Duration [hour]	FBHP @ pump intake expectation [bar]	Q_water expectation [m3/h]	Q_water cum. expectation [m3]
1	Production	100	3	170	4	12
2	Buildup	0	3	180	0	12
3	Production	300	3	128	16	60
4	Buildup	0	24	280	0	60

Table 2—Production test Slochteren - Mid Case

Step	Description	Injection pressure [bar]	Duration [hour]	FBHP @ pump intake expectation [bar]	Q_water expectation [m3/h]	Q_water cum. expectation [m3]
1	Production	100	3	174	16	48
2	Buildup	0	3	180	0	48
3	Production	300	3	156	66	246
4	Buildup	0	24	180	0	246

Table 3—Production test Slochteren - High Case

Step	Description	Injection pressure [bar]	Duration [hour]	FBHP @ pump intake expectation [bar]	Q_water expectation [m3/h]	Q_water cum. expectation [m3]
1	Production	100	3	179	35	105
2	Buildup	0	3	180	0	105
3	Production	300	3	178	98	399
4	Buildup	0	24	180	0	399

After an extensive hydraulic analysis a 2 3/8" CT string was ultimately selected, providing an optimal balance between internal flow capacity and operational logistics, while maintaining compatibility with standard transport configurations. This not only ensured adequate hydraulic performance for the jet pump across the anticipated production envelope, but also offered a clear advantage in terms of rapid mobilization between wells within the country, avoiding the need for specialized handling or oversized transport solutions. However, the string length was deliberately kept limited to minimize frictional pressure losses, which would otherwise raise circulating pressures and reduce the available jet pump intake flow. This design constraint directly influenced the achievable drawdown and was factored into the performance envelope to guarantee reliable operation across the defined production scenarios.

However, the jet pump configuration required, as will be detailed in the next dedicated BHA specifications section, both formation isolation and a controlled shut-in sequence to preserve reservoir data quality. For this purpose, a packer assembly incorporating a J-slot mechanism was considered. This design, however, relied on mechanical manipulation and a certainty set-down force to compress the rubber sealing elements adequately, ensuring proper anchoring and pressure integrity. Given the packer dimensions, a sustained axial compressive force at depth was essential to maintain element sealing throughout the entire production test.

To validate this approach, a dedicated tubing force analysis was also essential to confirm the achievable downhole set-down force with the 2 3/8" string under the expected operational conditions. In order to maximize the available compressive force, the CT string was deployed in a non-standard, reversed

orientation, positioning the section with greater wall thickness at the free end. This configuration provided an additional ~2,500 lbs of weight-on-bit (WOB) compared to the conventional setup, purely from the increased mass of the heavier-wall section, thereby enabling higher compressive loads during the packer setting sequence and improving sealing reliability. A significant operational advantage was provided by the real-time BHA telemetry (see previous HCTT section), which not only ensured accurate depth correlation and packer positioning, but also enabled continuous WOB monitoring throughout the production test. This capability ensured that the sealing elements remained engaged and fully effective, mitigating the risk of leak paths during dynamic flow conditions.

With the most complex and uncertain aspect of the well testing phase resolved, operational planning advanced to the detailed execution of the first phase, structured as a series of sequential runs. Each run was engineered to fulfill a specific operational objective while safeguarding data integrity and maximizing the quality of geothermal reservoir measurements.

First Run - Dummy Run, Temperature Logging, and Correlation. The initial operation served as a calibration and verification stage. A full dummy run was performed to confirm full wellbore accessibility to TD after drilling and completion activities. This pass also included a real-time temperature log, critical for establishing the baseline thermal profile of the well. The temperature data not only provided an early indication of geothermal gradient but were also essential for initiating cement testing in preparation for the decommissioning phase. Simultaneously, a gamma ray (GR) log was correlated with the Active Casing Collar Locator (ACCL) readings, the main sensors which was then used for the other operations as part of the HCTT RT BHA ensuring precise depth referencing for all subsequent runs. Due to the well configuration and the step created at the transition between the casing strings, a centralizer was installed at the bottom of the BHA. This ensured easier access to the target zones and prevented weight-on-bit transfer onto a WL BHA, which has lower compressive strength compared to a CT assembly.

Second Run - Reverse Circulation. The second run was preparatory in nature, designed to establish the appropriate underbalanced conditions prior to the planned TCP perforation. To maximize the value of the in-country real-time technology, the sensing BHA was retained in the downhole string, with only the double-flapper check valves removed. This preserved full telemetry and downhole diagnostics while maintaining critical safety features, including the burst disk and the ball-drop emergency disconnect system.

Reverse circulation had to be executed by bullheading nitrogen down the annulus (casing-CT) while simultaneously recovering wellbore fluids through the cable-equipped coiled tubing. This configuration enabled precise control of the hydrostatic pressure regime acting against the reservoir and improved monitoring of the downhole fluid response, ensuring that the well was consistently maintained in the required underbalanced condition.

Third Run - Tubing-Conveyed Perforating (TCP). Following calibration, the second run deployed more than 100 meters of perforating guns via coiled tubing, supported at surface by a custom-designed lifting plate capable of safely handling the combined weight of the TCP string. The perforating BHA was connected to the real-time telemetry system via a CARSAC interface, allowing continuous monitoring of annular and internal CT pressures during run-in-hole to avoid premature activation. Accurate depth positioning was ensured by the ACCL system, previously "calibrated" with the GR. The perforating guns were initiated using a pressure-activated firing head equipped with a bleed-off (weep) port to prevent closed-chamber between the MHA flapper check valve, which are normally closed spring loaded valves and the firing head, pressure build-up, an effect that could occur as wellbore fluids heated inside the CT, potentially causing unintended activation.

Fourth Run - Jet Pump Production Testing and Downhole water sample. The third run introduced the custom-engineered downhole jet pump assembly, designed specifically for geothermal well testing. The system enabled controlled production testing by lifting formation water using surface-supplied water as

the motive fluid, creating a closed-loop system that recycled part of the produced fluid to power the jet pump. This approach replicated the geothermal production conditions which are used to produce through a lift pump, providing representative performance data. The BHA was equipped with a memory-based water sampling tool, programmed with a delayed activation timer to collect fluid samples only at the end of the production phase, ensuring that the captured sample was formation-representative and uncontaminated by transient startup fluids.

Fifth Run - Real-Time Production Logging Tool (PLT). To further characterize reservoir performance, a real-time PLT run was conducted in injection mode. Powered via the copper conductor of the HCTT, the PLT delivered continuous downhole measurements without reliance on batteries, removing the operational time constraints typically associated with memory tools. Injection profiling was used to identify the most permeable intervals and to prioritize zones for potential future development. The real-time data stream allowed instant interpretation at surface, enabling operational adjustments during the run.

X-LOTS Engineering Phase

Another technically demanding component of the campaign was the execution of multi-layer MiniFrac/X-LOTS in the lower previously perforated formation, combined with a dedicated X-LOT on the overlying cap-rock. From a geomechanical standpoint, one of the principal objectives of the SCAN program was the in-situ determination of the minimum horizontal stress (S_{hmin}) at multiple depths within the target reservoir. These measurements are critical for optimizing stimulation designs, calibrating and validating local stress models, and quantifying intra-reservoir stress variability, particularly important in heterogeneous lithologies such as the Permian sandstones encountered in this case.

The cap-rock X-LOT addressed a complementary, yet equally critical objective: the direct determination of the mechanical integrity and fracture initiation pressure of the sealing formation. Such data are fundamental for confirming the cap-rock's ability to provide longterm hydraulic containment—a key parameter in subsurface risk assessments for geothermal field development, resource sustainability, and potential future re-injection or storage applications.

Achieving these objectives under fully rig-less conditions, while avoiding multiple deployments that would have rendered the scope economically unsustainable, required the development of a testing methodology capable of executing multiple, controlled X-LOTS within a single CT run. The adopted strategy comprised the execution of at least four discrete X-LOTS across perforated reservoir intervals during a single run-in-hole (RIH), immediately followed by a dedicated X-LOT on the cap-rock. For the latter, a different isolation system was deployed to both optimize the number of runs and ensure comprehensive testing of the entire cap-rock section, previously perforated through dedicated TCP operation.

For the reservoir section, a bespoke CT-conveyed inflatable straddle packer system was engineered, derived from an existing baseline design but extensively customized to meet the operational constraints of the well. These modifications were driven by the inherent challenges of operating in a pre-perforated casing environment, where exposed perforation tunnels present a significant risk of rubber element puncture, partial extrusion, or catastrophic sticking upon deflation, which, in the worst-case scenario, could result in a CT string becoming stuck in hole. The engineered configuration incorporated reinforced elastomer elements with enhanced extrusion resistance, and a mechanically actuated inflation/deflation sequence fully integrated with the HCTT technology. This design enabled reliable switching between inflation and deflation solely through controlled axial movement of the CT, eliminating the need to drop activation balls and thereby reducing common risks associated with inflatable packers, particularly unwanted or premature activation under fluctuating hydraulic conditions. The system's activation process also incorporated an integrated reservoir shut-in function, enabling immediate capture of post-injection pressure build-up data, recorded by two memory gauges housed within a custom-engineered gauge carrier placed between the rubber elements, ensuring redundancy and data integrity throughout the testing sequence. Although the use of inflatable

packers carried inherent limitations in terms of multi-cycle inflation endurance and potential vulnerability in perforated environments, the assessed risk-to-reward ratio was considered acceptable in view of the high-value stress data objectives and the operational efficiency afforded by a single-run, multi-zone testing approach.

For the cap-rock X-LOT, a different isolation strategy was implemented. A dedicated bridge plug was set above the reservoir to fully seal off the assessed pay zone, ensuring complete hydraulic isolation from the overlying cap-rock to be evaluated independently. Above the bridge plug, a cement retainer was installed following a dedicated TCP run to perforate the cap-rock. This configuration provided both a high-integrity mechanical barrier and a stable platform for subsequent decommissioning activities, thereby streamlining operations and optimizing both the X-LOT and the subsequent abandonment phase of the fully assessed zones below, where a redundant mechanical barrier and platform for the cement plug would have been required. The cement retainer was fitted with a custom-built gauge carrier accommodating redundant memory gauges for downhole pressure recording.

One significant engineering constraint was the unconventional tool size for CT deployment, which rendered conventional ball-drop activation impractical. Within the 2 3/8-in CT string housing an internal cable, it was not possible to drop from surface the activation ball (>1 in). This limitation was overcome by pre-positioning the ball directly onto the ball seat at surface and incorporating a weep-hole/circulation port in the setting tool, below the ball seat. This arrangement equalized internal and external hydrostatic pressures during run-in-hole, preventing premature activation.

The integration of HCTT technology was critical to the success of both X-LOT methodologies, reservoir and cap-rock, by providing precise depth control, real-time pressure monitoring, and accurate tool activation sequencing in a fully rig-less setup. Furthermore, redundant downhole memory gauges housed in custom-engineered BHAs safeguarded high-fidelity data acquisition, ensuring that fracture initiation and closure pressures could be reliably determined even in the event of a telemetry failure.

This integrated approach, combining a custom-engineered inflatable straddle packer system for the reservoir section with a mechanically robust bridge plug and cement retainer assembly for the cap-rock, enabled the SCAN project to acquire a multi-depth, high-fidelity in-situ stress dataset entirely under rig-less conditions. The methodology maximized information density per intervention while minimizing operational risk, cost, and environmental footprint, representing a step-change in the execution of multi-layer X-LOTs for geothermal developments. With the critical geomechanical objectives addressed through this dual-system strategy, the remainder of the operational design sequence could be defined and executed through the following runs.

Sixth Run - Straddle Packer Mini-Frac / X-LOT Testing. The sixth run involved the deployment of the previously described custom mechanical straddle packer assembly, designed for multiple inflation and deflation cycles. The system was required to be set multiple times, with a minimum target of four. Inflation was mechanically activated by a short axial movement of the CT (a few inches), thereby minimizing reliance on hydraulic sequencing delays. In addition, tailored operational procedures were implemented to mitigate the risk of the packer becoming lodged within open perforations.

Seventh Run - Bridge Plug Deployment for Caprock X-LOT. Following completion of reservoir testing, the seventh run consisted of the deployment of a large bridge plug to isolate the reservoir section. This isolation was required to proceed with the final X-LOT of the caprock interval, ensuring that formation integrity testing of the seal unit could be carried out without influence from the underlying reservoir.

Eighth Run - TCP to perforate Caprock. The eighth run was dedicated to deploying the TCP assembly to perforate the specific interval required for cap-rock testing. Given the criticality of this operation, precise depth control was paramount to ensure that perforations were accurately placed across the cap-rock section. Any deviation in depth could have compromised both the integrity of the subsequent X-LOT measurements

and the representativeness of the acquired geomechanical data. The integration of the real-time BHA and CCL correlation ensured accuracy in depth positioning, providing the operational assurance necessary to safely and effectively prepare the cap-rock for stress testing.

Nineth Run - Cement Retainer and X-Lot. The ninth run, dedicated to the planned cap-rock test, employed a cement retainer to provide upper-zone isolation and to enable evaluation of the entire cap-rock interval perforated during the earlier TCP run. Pressure response data were acquired by memory gauges. In addition, the retainer established a solid mechanical base and ensured a smooth transition for the placement of the first cement plug, marking the initial step of the decommissioning program for the fully assessed lower zone.

The ninth run marked both the final stage of the X-LOT program in the target intervals and the initiation of the decommissioning phase.

Decommissioning Engineering Phase

The decommissioning phase was, in principle, comparable to a conventional cementing operation. However, the integration of a hybrid conductor cable within the coiled tubing introduced non-standard constraints that required dedicated engineering considerations. Industry practice shows that several Operators and service providers impose restrictions on cement placement through cable-equipped CT strings. The main concerns are associated with the inability to displace a wiper dart to fully clean the CT bore after cement pumping, the risk of residual cement deposits impairing string functionality partial settling of cement particles within the CT string, potential solidification around the embedded conductor cable, and even full-bore blockage in worst-case scenarios, and the potential for axial cable movement toward the free end under the high-density slurry column. Such effects could compromise the long-term integrity of both the telemetry-enabled BHA and the conductor itself.

These risks were particularly relevant in the context of a multi-stage decommissioning program, where several cement plugs were interspaced with full pay-zone evaluation operations, as illustrated in [Figure 4](#). The repetitive nature of these operations further emphasized the need for a reliable and cable-compatible cementing methodology that would ensure both effective zonal isolation and preservation of the ICT string's operational integrity throughout the campaign.

Replacing the CT string with a conventional cable-free CT for each cementing stage, together with the repeated rig-up/rig-down operations this would have required, was deemed operationally, economically, and safety-wise unsustainable. The additional nonproductive time, cumulative handling risks, and logistical complexity would have significantly increased overall project exposure. Furthermore, the environmental impact of such an approach could not be overlooked. Extended operational durations and the mobilization of additional equipment would have translated directly into higher greenhouse gas emissions. Based on standardized CO₂ emission factors (ref. [SPE-224111](#)), the adopted rigless, cable-compatible approach resulted in an estimated saving of approximately 30 metric tons of CO₂ compared with a conventional multi-string replacement strategy, an important consideration within the framework of green initiatives and sustainability objectives underpinning the project.

Given the scope of work, it was imperative to execute the decommissioning phase using the equipment already mobilized on site, while at the same time preserving the integrity of the hybrid coiled tubing technology and ensuring full functionality of the downhole realtime BHA. Efficient cement placement had to be achieved without compromising either the conductor cable or the CT string. As shown in the completion diagrams ([Figures 1 and 4](#)), three target zones were identified for sequential evaluation and subsequent decommissioning, alternated with other operational phases. This cyclic execution imposed the need for a cementing methodology that could be reliably repeated multiple times without leaving residual cement within the lower CT coils, where cleaning and retrieval are impractical, and without imposing mechanical or chemical damage to the cable. Ensuring repeatability, string integrity, and downhole tool survivability therefore became central design criteria in developing a suitable cementing solution for this campaign.

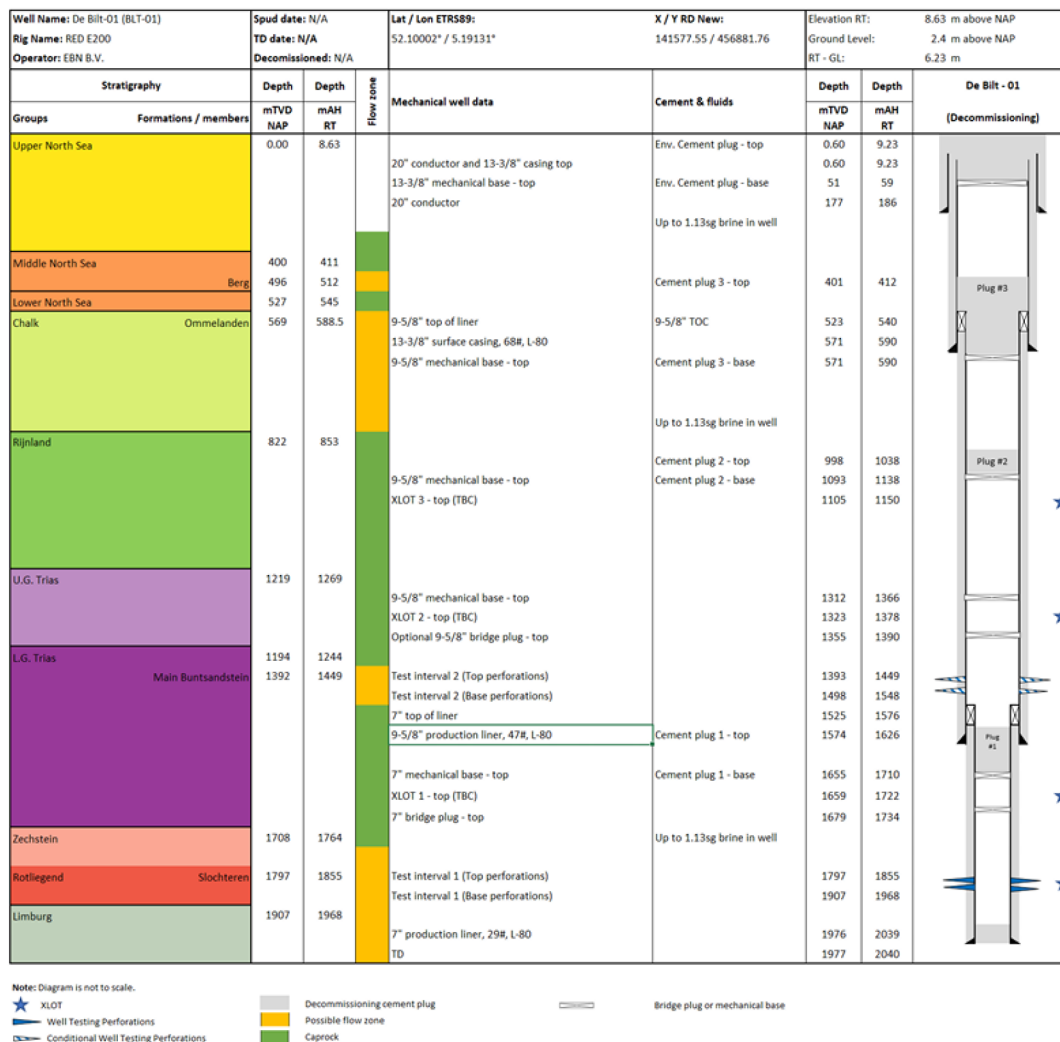


Figure 4—Final Well Status

From a hydraulic standpoint, additional constraints emerged due to the requirement of maintaining pump rates > 3.0 bbl/min. This threshold was dictated by the large-diameter CT string selected for the well testing phase, as hydraulic simulations demonstrated that flow rates below this value could trigger free-fall phenomena when pumping slurries in the density range of 1.80-1.90 SG. Standard API Class G cement was therefore considered unsuitable, as its coarse particle size distribution and high solids loading presented multiple risks. Most critically, the coarse particulate nature of Class G cement significantly increased the risk of erosion damage to the real-time BHA, which contained reduced-ID flow paths. Finite Element Analysis (FEA) and fluid dynamics modeling confirmed that the BHA could tolerate pumping rates up to 4.0 bbl/min with solids-free fluids, but only up to 2.0 bbl/min for solids-laden slurries. This limitation was incompatible with the minimum flowrate required to avoid free-fall phenomenon, effectively ruling out the use of conventional coarse Class G cement in this application.

To address these challenges, a microfine cement system was selected. Microfine cements, characterized by particle sizes typically < 10 μm and Blaine fineness values exceeding $8,000$ cm^2/g , provided a tailored solution to the operational constraints. The microfine system delivered several key advantages in this context:

- Near-solid-free hydraulic response: The ultra-fine particle size distribution generates a slurry with significantly reduced abrasive potential, mitigating erosion risks to the sensitive flow paths of the real-time BHA.
- Cable integrity preservation: The homogeneous rheology and lower hydrodynamic forces of the microfine slurry reduced the risk of cable displacement toward the CT free end, a known concern during high-density fluid pumping.
- Reduced residue accumulation: Unlike Class G systems, microfine slurry minimizes the probability of coarse particulate deposition within the lower CT coils, where cleaning and retrieval are impractical.

Laboratory characterization of the selected microfine system confirmed a particle size distribution between 1 and 30 μm , with a median (d₅₀) of 5-10 μm and no particles above 30 μm . The ultrafine grading, produced through controlled grinding and classification, yields a specific surface area (Blaine fineness) exceeding 8,000 cm^2/g , compared with 3,000-3,500 cm^2/g for API Class G cement. When hydrated, the system develops a highly alkaline environment (pH 11-13.5), producing dense hydrates with very low permeability. Under simulated geothermal brine flow, the cement achieved a Permeability Reduction Index (PRI) $>10^5$, an order of magnitude higher than Class G formulations. Operationally, the absence of coarse particles eliminates the risk of sedimentation or bridging in the CT string and minimizes abrasion within the reduced-ID passages of the real-time BHA, ensuring compatibility with the rig-less hybrid CT methodology while still providing long-term sealing performance.

From a barrier-performance standpoint, this enhanced fineness delivers ancillary benefits beyond the hydraulic compatibility requirements of the CT-cable system. In particular, the finer particle size markedly decreases matrix permeability after set. Laboratory testing of the selected microfine cement system demonstrated a Permeability Reduction Index (PRI) exceeding 10^5 against brine flow under simulated geothermal conditions, compared with typical values of 10^3 - 10^4 for Class G slurries. This improved sealing efficiency, while not critical to the abandonment objectives in this case, nonetheless represents an additional value in terms of long-term integrity and zonal isolation.

Rheologically, the selected microfine cement system exhibited lower yield point and plastic viscosity compared to equivalent-density Class G formulations. This enabled stable placement at rates up to 3.0 bbl/min through the 2 3/8-in CT string without generating excessive friction pressures. Such characteristics made microfine cement particularly well suited for the SCAN decommissioning program, where plugs had to be repeatedly placed in rigless mode using a cable-equipped Hybrid CT string. The slurry's fine particle size distribution minimized risks of internal abrasion and premature cable wear, while also preventing erosion damage to the real-time BHA components, thus preserving system integrity across multiple cycles.

The combination of favorable rheological behavior, reduced abrasivity, and high sealing performance allowed cement plugs to be efficiently placed while maintaining full CT and BHA operability. This approach ensured that the same HCTT-equipped CT string could be utilized seamlessly across diagnostic, intervention, and decommissioning phases, eliminating the need for string replacement and thereby reducing mobilization requirements, operational downtime, and overall project cost.

With this key challenge addressed, the operational sequence could have been planned with the below additional runs for the decommissioning phase.

Ninth Run - Cement Plug. As part of the ninth run, once the X-LOT sequence had been concluded using the cement retainer and the subsequent sting-out completed, a cement plug was placed above the retainer to establish a permanent barrier in accordance with the requirements of the local mining authority.

Tenth Run - Cement plug verification. The tenth run concluded the decommissioning phase by verifying the placement and top of the cement plug, and by performing a 1-ton weight test in compliance with local mining authority regulations to confirm barrier integrity. In the same run, operations were transitioned

seamlessly into the next abandonment stage, with underbalanced conditions established in preparation for the TCP perforation of the overlying pay zone. This dual-purpose approach optimized operational efficiency by combining plug verification and preparation for the subsequent test interval within a single CT deployment.

Bha Specifications

BHA Run 1 - Dummy Run, Temperature, CCL and GR Log. The first BHA deployed was equipped with a cablehead and configured as a conventional wireline logging string in order to leverage the superior accuracy of quartz-based temperature and pressure sensors and to establish a precise depth correlation through the combined use of CCL and GR measurements. A bottom centralizer was also included in the assembly to ensure smooth transition across each casing diameters, from the 13%-in section through the 9%-in and into the 7-in liner.

BOTTOM HOLE ASSEMBLY DETAILS						
	Schematic	Item	Description	Tool Connections	Tool OD (in)	Tool ID (in)
1		1	External Slip Connector	Coiled Tubing 2-3/8" PAC pin	3,125	1,38
		2	X-Over	2" 3/8" PAC box 1,5" AMMT	2,875	1,00
2		3	Straight Bar	1,5" AMMT 1,5" AMMT	2,125*	1,00
		4	Cable Head with Circulation X-over and Flapper CV	1,5" AMMT WL Connection	2,125*	N/A
3		5	WL Sensing BHA with (from top to bottom): - Crossover Ultrawire Toolbus to Ultralink - Knuckle Joint - Quartz Pressure - Collar Locator - Production Gamma Ray - Knuckle Joint - Capacitance Temperature Flow	WL Connection	1,69*	N/A
4		6	Roller Centralizer	WL Connection	1,69*	N/A
		7	Bullnose	WL Connection	1,69*	N/A
5						
6						
7						
		BHA Total Length (m)				7,48
		BHA Prepared By				Massimiliano Gaeta

Figure 5—First BHA

The quartz transducers provided the required stability and fidelity for geothermal applications, with pressure resolution typically better than 0.01 psi and temperature resolution finer than 0.001 °C. This capability was critical for establishing the initial baseline dataset and for calibrating subsequent runs.

The inclusion of the CCL, combined with gamma ray response, allowed accurate correlation between the completion and the reservoir for later perforating and testing operations.

BHA Run 2 - Reverse Circulation. The second BHA incorporated the Real-Time Sensing assembly described in the HCTT specifications, adapted to allow direct fluid intake from the wellbore by removing the downhole flapper check valve. In this configuration, the string was capable of safely admitting wellbore fluids during reverse circulation operations, thereby maintaining a controlled underbalanced condition while providing continuous real-time telemetry of annular and tubing pressures and temperatures. A bottom centralizer was retained in the assembly to ensure smooth transition across the step-down casing diameters.

BOTTOM HOLE ASSEMBLY DETAILS						
Schematic	Item	Description	Tool Connections	Tool OD (in)	Tool ID (in)	Tool length (m)
1	1	External Slip Connector	Coiled Tubing 2-3/8" PAC pin	3,125	1,38	0,33
	2	X-Over	2" 3/8 PAC box 2" - 8 SA pin	2,875	1,5"	0,15
2	3	Cable Head	2" - 8 SA box 2" - 8 SA pin	2,375	0,55	0,56
3	4	Real Time Sensing BHA: - P&T Int & Ext - Tension & Compression - CCL - Vibration - Inclination - TOB	2" - 8 SA box	2,375	0,55	1,75
			1,5" AMMT box		1,00	0,30
4	5	X-Over	1,5" AMMT box 2 3/8" PAC pin	2,875	1,00	0,30
	6	Ball Operated Shear Sub - Disc Ball: 1/2" w/ 5K psi BD	2 3/8" PAC box 2 3/8" PAC pin	2,875	0,50	0,56
	7	Centraliser Body, for CS Centralizer Sleeve Interchangeable	2 3/8" PAC box 2-3/8" PAC pin	7,000	1,00	0,40
5	8	Fullbore nozzle	2 3/8" PAC box	2,875	2"	0,3
6						
7						
8						
		BHA Total Length (m)		4,65		
		BHA Prepared By		Massimiliano Gaeta		

Figure 6—Second BHA

BHA Run 3 - TCP. The third BHA was configured for the TCP perforation run and consisted of the Real-Time Sensing assembly coupled with the CARSAC connector and a pressure-activated firing head, deployed above a long TCP gun string. This configuration enabled continuous monitoring of wellbore parameters during run-in-hole, ensuring safe deployment and accurate depth control of the perforating assembly, while the pressure-activated firing head provided a reliable initiation mechanism that was fully compatible with the CT conveyance system.

BOTTOM HOLE ASSEMBLY DETAILS							
Schematic	Item	Description	Tool Connections	Tool OD (in)	Tool ID (in)	Tool lenght (m)	
1		1	External Slip Connector	Coiled Tubing	3,125	1,38	0,33
			2-3/8" PAC pin				
2		2	X-Over	2" 3/8" PAC box	2,875	1,5"	0,15
			2" - 8 SA pin				
3		3	Cable Head	2" - 8 SA box	2,375	0,55	0,56
			2" - 8 SA pin				
3		4	Real Time Sensing BHA: - P&T Int & Ext - Tension & Compression - CCL - Vibration - Inclination - TOB	2" - 8 SA box	2,375	0,55	1,75
				1,5" AMMT pin			
4		5	X-Over	1,5" AMMT box	2,875	1,00	0,30
				2 3/8" PAC pin			
4		6	Ball Operated Shear Sub - Disc Ball: 1/2" w/ 5K psi BD	2 3/8" PAC box	2,875	0,50	0,56
				2 3/8" PAC pin			
5		7	X-Over	2 3/8" PAC box	2,875	1,00	0,30
				Regular pin			
5		8	CARSAC	Regular box	3,125	1,50	0,90
				Regular pin			
6		9	Centraliser Body, for CS Centralizer Sleeve Interchangeable	2 3/8" PAC box	7,000	1,00	0,40
				2-3/8" REG pin			
6		10	X-over	Regular box	2,875	TBC	0,20
				2 7/8" EU PIN			
7		11	PAFH	2 7/8" EU BOX	4,500	N/A	0,35
				W/L Thread			
7		12	TCP	W/L Thread	4,500	N/A	TBC
8							
9							
10							
11							
12							
							
							
							
							
							
							
							
							
							
							
							
							
							
							
							
							
							
							
							
							
							

Figure 7—Third BHA

BHA Run 4 - Production Test and Downhole Sampling. One of the most critical and heavily engineered BHAs of the campaign was the production testing assembly, previously referenced, which incorporated a downhole jet pump system specifically designed for coiled tubing deployment. The assembly was complemented by redundant memory gauges positioned below the isolation element to capture high-resolution reservoir pressure transients immediately following shut-in, thereby eliminating storage effects and ensuring accurate reservoir characterization. In addition, a downhole water sampling tool was integrated into the same run, preprogrammed to collect formation fluid only after the production phase had been completed, thus guaranteeing representative samples. Figure 8 presents a schematic of the complete production test BHA.

BOTTOM HOLE ASSEMBLY DETAILS						
Schematic	Item	Description	Tool Connections	Tool OD (in)	Tool ID (in)	Tool length (m)
1	1	External Slip Connector	Cooled Tubing	3,125	1.38	0.33
			2-3/8" PAC pin			
2	2	X-Over	2" 3/8" PAC box	2,875	1.50	0.17
			1.5" AMMT pin			
3	3	Cable head	1.5" AMMT box	2,375	0.75	0.45
			1.5" AMMT pin			
4	4	X-Over	1.5" AMMT box	2,875	1.00	0.20
			2-3/8" PAC pin			
5	5	Dual Circulation Sub	2-3/8" PAC box	2,875	1.25	0.30
			2-3/8" PAC pin			
6	6	Ball Operated Shear Sub - Disc Ball: 1/2"	2-3/8" PAC box	2,875	0.50	0.56
			2-3/8" PAC pin			
7	7	Centralizer Body, for CS Centralizer Sleeve Interchangeable	2-3/8" PAC box	7,000	1.00	0.40
			2-3/8" REG pin			
8	8	X-Over	2-3/8" PAC box	3,125	1.00	0.30
			2-3/8" REG pin			
9	9	CARSAC	2-3/8" REG box	3,125	1.50	0.52
			2-3/8" REG pin			
10	10	X-Over	2-3/8" REG box	3,125	1.00	0.30
			2-3/8" PAC pin			
11	11	X-Over	2-3/8" PAC box	3,125	1.36	0.30
			3 1/2" EUE pin			
12	12	Pump Joint	3 1/2" EUE box	3,500	2.90	1.49
			3 1/2" EUE box			
13	13	Jet Pump Housing and Outlet	3 1/2" EUE box	5,000	N/A	0.91
			SA			
14	14	Check Valve Assembly	SA	5,880	N/A	0.62
			3 1/2" IF pin			
15	15	Auto-J Coil Operated Compression Packer	3 1/2" IF box	8,250	2.50	1.51
			3 1/2" IF pin			
16	16	Gauge Carrier and Memory P&T	3 1/2" IF box	5,125	2.50	0.67
			3 1/2" IF pin			
17	17	X-Over	3 1/2" IF box	3,125	2.50	0.30
			3 1/2" EUE pin			
18	18	Pump Joint Intake with bull nose	3 1/2" EUE box	3,500	2.90	1.49
			3 1/2" EUE box			
19	19	Wireless gauges	1-1/16 SR	1,500	N/A	2.00
20	19	WL Memory Sampling BHA Assy with: - Knuckle Joint - PDS Clock Housing - PDS Air Chamber - SPS Sample Chamber - Knuckle Joint - PDS Clock Housing - PDS Air Chamber - Chamber - SPS Sample Chamber - Centralizer - Bellowse	15/16" SR	1,68"	N/A	10.68
14						
15						
16						
17						
18						
19						
BHA Total Length (m)			23.50			
BHA Prepared By			Massimiliano Gaspari			

tubing strings. The packer had an outer diameter of 8.25 in., optimized for deployment inside 9% in., 47 lb/ft casing, providing secure zonal isolation during flow and shut-in phases.

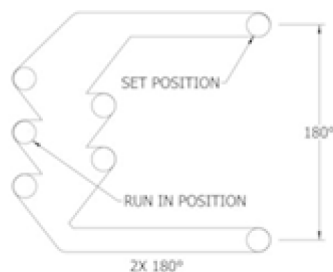


Figure 9—J-Latch

The pump operated on the Venturi principle, adapted from concentric CT applications but redesigned for deployment with a standard 2 $\frac{3}{8}$ -in CT string. High-pressure motive water was pumped down the CT and discharged through a converging nozzle inside the downhole pump body, generating a localized low-pressure zone at the throat. This pressure drop induced the inflow of reservoir fluids into the pump intake. Within the throat, the high-velocity motive stream and the formation inflow mixed under strongly turbulent conditions, where momentum transfer occurred through shear forces and turbulent eddies. Although this mixing process is characterized by energy dissipation (so-called throat-mixing losses), the geometry of the nozzle-throat-diffuser system was optimized to maximize efficiency and maintain a stable lifting regime. The mixed fluid stream then entered the diffuser section, where velocity was gradually reduced and kinetic energy was reconverted into pressure, providing sufficient lift to overcome the hydrostatic head of the annular column, the frictional pressure losses in the CT-casing annulus, and the imposed surface backpressure.

Because the packer provided hydraulic isolation, the combined stream of motive water and produced formation fluid was forced to flow exclusively into the CT-casing annulus, ensuring controlled production and accurate surface metering. By tailoring the nozzle and diffuser configurations, the system was able to operate across a wide envelope, from low drawdown to high-flow conditions, with stable performance confirmed through hydraulic simulations. Flow modeling demonstrated operational stability between 0.5 and 3.5 m³/min ($\approx$ 4,500–31,700 bbl/d), covering the expected range of reservoir productivity. The use of water as the sole motive fluid ensured operational simplicity, avoided nitrogen use and logistic issue, and provided a representative simulation of permanent geothermal production pumping systems.

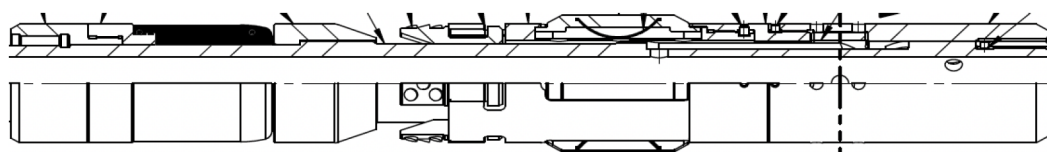


Figure 10—Isolation Packer Assembly

A dual-flapper shut-in valve was integrated immediately above the packer, enabling rapid reservoir isolation within seconds of ceasing pumping operations. This feature was essential to eliminate wellbore storage effects observed in earlier SCAN wells and to capture high-fidelity pressure transients directly from the reservoir. The capability to toggle from production to shut-in without retrieval of the CT string represented a major advancement in operational efficiency and data quality. To ensure robust data capture, the BHA integrated a dual-gauge strategy: high-precision quartz memory gauges, deployed below the packer, recorded bottomhole pressure and temperature during shut-in, unaffected by hydrostatic column interference, while wireless real-time gauges positioned within the CT assembly provided surface readout

of pressure and flow dynamics during production, enabling operators to adapt pumping parameters in real time. The combination of real-time and memory datasets allowed continuous optimization of test procedures and provided redundancy to guarantee dataset integrity.

The downhole memory sampling system deployed beneath the jet pump assembly was a pressure-compensated single-phase sampler (PCS), designed to preserve fluid integrity under high-pressure, high-temperature geothermal conditions. The sampler consisted of a 560 cc sample chamber hydraulically balanced against a 385 cc nitrogen precharge chamber, enabling full pressure compensation across the sampling cycle. The pressure-compensation mechanism ensured that shrinkage effects—typically ~15% of chamber volume during retrieval and cooling, was offset by a controlled nitrogen cushion, preserving the single-phase character of the captured sample. The sampler pre-charge was calculated from the known bottomhole pressure using the relation:

$$P_1 = 1.1 \times P_2 \times V_2 / V_1$$

with nitrogen charging pressure subsequently derived as

$$P_N = P_1 \times V_1 / V_N.$$

Mechanically, the PCS incorporated an electronic clock and trigger mechanism, allowing timed closure at reservoir depth to ensure the captured sample was representative of stabilized production conditions. The system also included a pressure safety relief valve (PSRV) and a constant flow regulator, providing operational redundancy and ensuring controlled closure dynamics even under variable reservoir drawdown.

This tool design allowed high-fidelity reservoir fluid characterization to be integrated into the same run as the jet pump production test, ensuring that samples were taken only at the end of the stabilized flow period, immediately after shut-in, and directly below the isolation packer, thus guaranteeing representative reservoir fluid capture for geochemical and scaling analysis.

BHA Run 5 - PLT/ Injection test. The fifth BHA was configured for the production logging (PLT) phase, aimed at evaluating zonal contribution during the injection tests and establishing a complete injection profile across the interval of interest. The assembly was run on coiled tubing with a cable head and a standard wireline PLT toolstring, deployed as illustrated in [Figure 5](#). To maximize diagnostic resolution, the toolstring was designed with a full-bore spinner, as no sand production or significant debris inflow was anticipated under injection conditions. The full-bore spinner configuration provided optimal sensitivity to low-velocity flow regimes, ensuring accurate quantification of zonal inflow contributions and good injection allocation analysis.

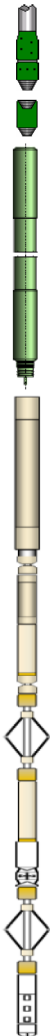
BOTTOM HOLE ASSEMBLY DETAILS						
Schematic	Item	Description	Tool Connections	Tool OD (in)	Tool ID (in)	Tool length (m)
	1	External Slip Connector	Coiled Tubing 2-3/8" PAC pin	3,125	1,38	0,33
	2	X-Over	2" 3/8 PAC box 1,5" AMMT pin	2,875	1,00	0,15
	3	Cable Head with Circulation X-over w Flapper CV	1,5" AMMT box WL Connection	2,125"	N/A	1,49
	4	WL Sensing BHA: - Crossover Ultrawire tool bus to Ultralink - Knuckle Joint - Quartz Pressure - CCL - GR - Knuckle Joint - Roller Centralizer - Inline spinner - Roller Centralizer - Capacitance Temperature Flow - Continuous Flow Meter	WL Connection	2,130	N/A	4,62
	5					
BHA Total Length (m)						6,59
BHA Prepared By						Massimiliano Gaeta

Figure 11—Fifth BHA

BHA Run 6 - Multilayers X-Lot with straddle assembly. The sixth BHA was the engineered straddle packer assembly designed for the multilayer X-LOT program. The system consisted of a dual-packer straddle configuration integrated with the HCTT real-time sensing BHA, providing depth control, telemetry, and downhole pressure monitoring.

BOTTOM HOLE ASSEMBLY DETAILS						
Schematic	Item	Description	Tool Connections	Tool OD (in)	Tool ID (in)	Tool length (m)
	1	External Dimple Connector for 2" CT	Coiled Tubing 2-3/8" PAC pin	2,875	1,50	0,19
	2	X-Over	2" 3/8 PAC box 2" - 8 SA pin	2,875	1,00	0,16
3	3	Cable Head	2" - 8 SA box 2" - 8 SA pin	2,375	0,55	0,56
	4	Real Time Sensing BHA: - P&T Int & Ext - Tension & Compression - CCL - GR - Vibration - Inclination	2" - 8 SA box 1,5" AMMT pin	2,375	0,55	1,75
4	5	X-over	1,5" AMMT pin 2, 3/8" PAC	2,875	0,50	0,35
	5	MHA assembly - without check valve	2 3/8" PAC	2,875	0,90	0,90
5	7	X-Over	2 3/8" PAC 2-3/8" REG	3,125	1,50	0,35
6	8	CARSAC	2-3/8" REG	3,125	1,50	0,52
	9	X-Over	2-3/8" REG 2 3/8" PAC	3,125	1,50	0,35
7	10	Straddle Assembly with multistage position and gauge carrier with P&T between packer elements.	2 3/8" PAC	5,5"	2,00	TBC
8						
9						
10						
BHA Total Length (m)						5,13
BHA Prepared By						Massimiliano Gaeta

Figure 12—Sixth BHA

The straddle system employed 5-in contra-wound inflatable high-pressure packer elements, each reinforced with four internal steel wire layers, rated to withstand injection pressures up to 10,000 psi while ensuring full retrievability. The heavy-duty rubber cover was designed to resist extrusion into perforation tunnels and to prevent sticking during deflation, even after multiple inflation/deflation cycles.



Figure 13—Reinforced Rubber Element

Key mechanical features included a balanced piston mechanism to eliminate tool movement during inflation, and a bypass vent system to avoid artificial pressure spikes by releasing wellbore fluid between the elements during packer setting. A shut-in valve positioned above the upper packer enabled clean reservoir isolation after injection, preventing column storage effects and ensuring accurate fall-off pressure data. For data acquisition, the BHA incorporated dual memory gauge carriers mounted between the packer elements, providing redundancy and decoupling fracture pressure monitoring from tool-body dynamics. Internal filtration modifications allowed compatibility with the fluids used for the mini-frac, and circulation ports permitted conditioning of the CT string before injection, ensuring clean frac fluids entered the formation.

BHA Run 7 - Bridge Plug. The seventh BHA was configured for the deployment of a large-diameter bridge plug, serving as a mechanical isolation barrier between the fully assessed reservoir interval and the overlying cap-rock section to be subsequently tested. To ensure accurate depth placement and mitigate the risk of inadvertently setting the plug across an untested interval of interest, the assembly was integrated with the HCTT downhole telemetry system.

BOTTOM HOLE ASSEMBLY DETAILS						
Schematic	Item	Description	Tool Connections	Tool OD (in)	Tool ID (in)	Tool length (m)
1	1	External Dimple Connector	Coiled Tubing 2-3/8" PAC pin	2,875	2,00	0,19
	2	X-Over	2" 3/8 PAC box 2" - 8 SA pin	2,875	1,50	0,16
3	3	Cable Head	2" - 8 SA box 2" - 8 SA pin	2,375	0,55	0,56
	4	Real Time Sensing BHA: - P&T Int & Ext - Tension & Compression - CCL - GR - Vibration - Inclination	2" - 8 SA box 1,5" AMMT pin	2,375	0,55	1,75
	5	X-Over	1,5" AMMT box 2 3/8" PAC	5,000	1,00	0,35
4	6	Motorhead assembly	2 3/8" PAC	2,125	0,75	0,90
	7	X-Over	2 3/8" PAC 2-3/8" REG	3,125	1,50	0,35
6	8	CARSAC	2-3/8" REG	3,125	1,50	0,52
	9	X-Over	2-3/8" REG 2 3/8" PAC	3,125	1,50	0,35
7	10	X-Over	2 3/8" PAC 3 1/2 IF	3,500	N/A	0,35
	11	Setting Tool	3 1/2 IF	3,500	N/A	2,44
8	12	Bridge Plug	Internal Connection	7,71" for the 9 5/8" or 6,8 for the 7" Liner	N/A	0,91
9						
10						
11						
12						
BHA Total Length (m)			8,83			
BHA Prepared By			Massimiliano Gaeta			

Figure 14—Seventh BHA

The bridge plug design featured high-expansion elastomer elements and a slip-based anchoring mechanism rated for high differential pressures, ensuring a secure seal across the casing ID. The telemetry-enabled deployment enhanced placement confidence and allowed continuous monitoring of setting parameters, ensuring the plug achieved full anchoring and pressure integrity before proceeding to the cap-rock X-LOT sequence.

BHA Run 8 - Caprock Perforation. The eighth BHA replicated the configuration of the third assembly. The only modification was the adjustment of the TCP gun string length to match the shorter interval designated

for perforation, while maintaining identical safety features and real-time monitoring capability to ensure accurate depth control and reliable gun activation.

Run 9 - Cement Retainer for X-Lot and P&A. The ninth BHA was configured with a cement retainer assembly, as illustrated in Figure 15. The retainer was selected for its dual functionality: enabling execution of the cap-rock X-LOT during the sting-in phase, and subsequently serving as a mechanical platform for cement plug placement during the decommissioning sequence. The setting tool was engineered with a large internal flow area (>1 in.), ensuring compatibility with cement placement operations and minimizing the risk of hydraulic restriction during pumping. The sting-in/sting-out capability allowed the assembly to be re-engaged after integrity verification of the retainer, providing operational flexibility and a secure transition from testing to abandonment activities within a single deployment.

BOTTOM HOLE ASSEMBLY DETAILS						
Schematic	Item	Description	Tool Connections	Tool OD (in)	Tool ID (in)	Tool length (m)
1	1	External Dimple Connector	Coiled Tubing	2.875	2.00	0.19
			2-3/8" PAC pin			
2	2	X-Over	2" 3/8 PAC box	2.875	1.50	0.16
			2" - 8 SA pin			
3	3	Cable Head	2" - 8 SA box	2.375	0.55	0.56
			2" - 8 SA pin			
3	4	Real Time Sensing BHA: - P&T Int & Ext - Tension & Compression - CCL - GR - Vibration - Inclination	2" - 8 SA box	2.375	0.55	1.75
			1.5" AMMT pin			
4	5	X-Over	1.5" AMMT box	5.000	1.00	0.35
			2 3/8" PAC			
5	6	Motorhead assembly - without check valve	2 3/8" PAC	2.125	0.75	0.90
			2 3/8" PAC			
5	7	X-Over	2 3/8" PAC	3.125	1.50	0.35
			2-3/8" REG			
6	8	CARSAC	2-3/8" REG	3.125	1.50	0.52
			2-3/8" REG			
6	9	X-Over	2 3/8" PAC	3.125	1.50	0.35
			2 3/8" PAC			
7	7	Setting Tool	2 3/8" PAC	3.600	TBC	2.45
8	8	Stinger	Internal Connection	N/A	TBC	0.81
8	9	Cement Retainer	Internal Connection	8.125" for the 9 5/8" or 6.8 for the 7" Liner	TBC	0.91
9						
10						
11						
12						
13						
14						
BHA Total Length (m)				9.31		
BHA Prepared By				Massimiliano Gaeta		

Figure 15—Ninth BHA

10 BHA - Cement Plug Verification. The cement plug verification run employed a configuration replicating BHA number 2. This assembly, incorporating the Real-Time Sensing BHA, which provided precise depth correlation to confirm plug position and continuous downhole telemetry during the operation. The integrated tension/compression module ensured accurate monitoring of weight-on-bit (WOB), allowing the prescribed 1-ton weight test on the cement plug to be executed with high fidelity. In addition to validating plug integrity, this run also functioned as a transitional step toward the evaluation of the second reservoir zone, by establishing the required underbalanced condition for the subsequent perforation of the upper interval.

Results

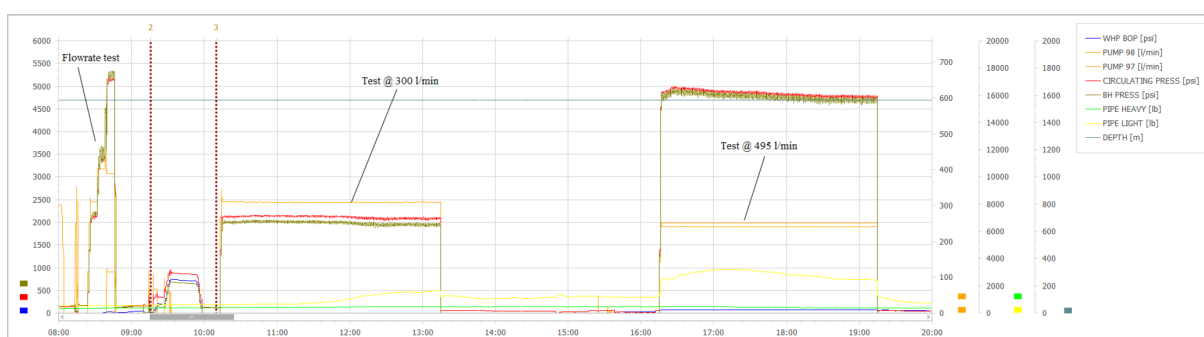
The execution confirmed the feasibility of conducting the entire operation in a fully rigless mode, while still delivering high-quality formation assessment data aligned with the project's objectives. Each individual run contributed incremental value to the evaluation program, and the main operational challenges were successfully mitigated as anticipated.

The initial run reached TD smoothly without operational complications, providing accurate Temperature log responses and reliable depth correlation through combined CCL and GR measurements.

The subsequent underbalanced operation, executed through controlled reverse circulation, demonstrated the value of combining realtime enabled BHA capabilities with a precision surface fluid-management strategy. This dual-layer monitoring framework, real-time annular pressure measured directly at the BHA, complemented by surface monitoring of fluid displacement, ensured rigorous validation of the designed underbalance conditions.

The most anticipated outcome of the intervention was the production test, which successfully delivered reservoir inflow in line with the previously established mid-case expectations. Remarkably, this was achieved at an intermediate injection rate of 495 L/min (~3.1 bbl/min), used as power fluid to activate the jet pump, rather than at the maximum simulated flow rate. The test thereby validated both the robustness of the engineered solution and the performance of the full assembly that effectively replaced the conventional rig.

An overview of the production test at the two main flow rates, separated by a ~3-hour reservoir shut-in, is shown in Figure 8. This sequence provided valuable pressure-transient information and fluid sampling under controlled conditions.



Graph 1—Jet Pump Run Overview

The engineered jet pump assembly delivered multiple operational advantages in a single deployment:

- Production testing under controlled flow conditions,
- Reservoir shut-in and transient logging,
- Representative fluid sampling, and

- Operational flexibility with minimal surface footprint and without the need for heavy rig equipment.

Furthermore, flow rates and drawdown sequences could be adjusted dynamically based on real-time telemetry, allowing operators to optimize test conditions on the fly. The assembly also featured immediate shut-in capability and redundant downhole pressure monitoring, ensuring data integrity and operational safety throughout the test.

Overall, the jet pump system enabled the SCAN team to replicate geothermal production conditions with an unprecedented level of control in a rigless environment, confirming its suitability for multi-zone data acquisition programs where rapid redeployment and efficient reservoir evaluation are critical.

Another anticipated milestone, arguably the one with the highest operational risk due to the potential for CT stuck-in-hole scenarios, was the execution of the multilayer mini-frac/XLOT program. In this case as well, the operation was successfully completed, with four mini-fracs performed in a single coiled tubing run without incident. The reinforced rubber elements of the straddle packer assembly withstood the damaged cased-hole conditions caused by pre-perforation, and the closed-loop packer setting/unsetting system functioned reliably, facilitating seamless transitions between test intervals. As illustrated in Figure X, only minor surface scratches were observed on the packer elements, confirming their mechanical resilience.



Figure 16—Rubber Element Post Run

The graphs in Figure X illustrate the packer's performance across the three critical stages of the operation: inflation, mini-frac execution, and deflation prior to repositioning. Each stage was completed smoothly, with transitions controlled by incremental axial movement of the coiled tubing—typically a few inches—between successive test intervals. These precise repositioning steps were continuously monitored and validated through the downhole real-time sensing BHA, ensuring accurate tool control, effective zonal isolation, and seamless progression from one stage to the next.

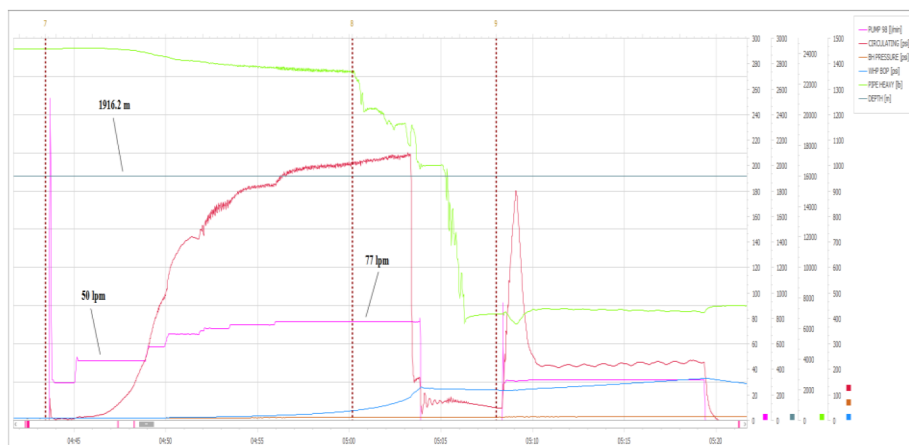


Figure 17—Inflation step.

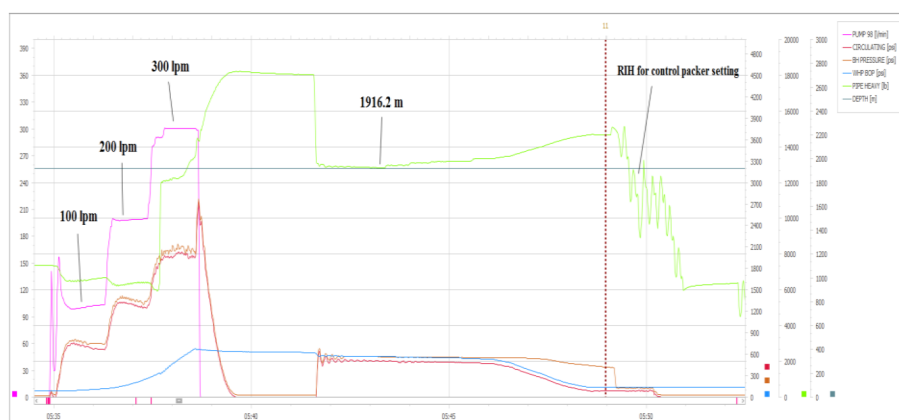


Figure 18—Mini-frac step.

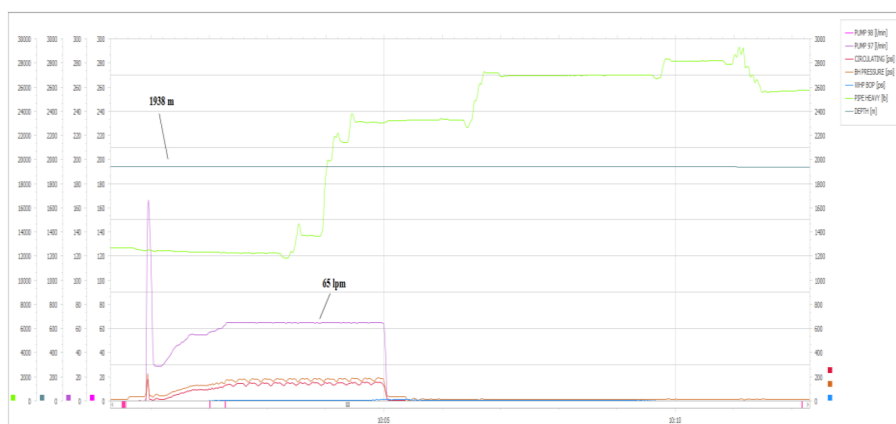


Figure 19—Deflation step.

An additional cap rock X-LOT was performed successfully followed by the cement placement above. The ability to carry out a sting-out and verify the cement retainer's integrity prior to initiating the cap-rock X-LOT, provided critical assurance that the test could be executed accurately and that the resulting stress data would be reliable. After verification, the string was re-stung into the retainer, enabling the caprock X-LOT sequence to be conducted under fully controlled conditions. Upon completion of the stress test, the cement retainer remained in place as an integrity barrier, creating a robust mechanical platform for the

subsequent cement plug. This configuration ensured that the first fully evaluated zone could be permanently abandoned while maintaining both operational efficiency and regulatory compliance.

The final cementing phase confirmed the overall robustness of the approach. Post-job verification demonstrated that the cementing fluids, even when pumped at high rates, did not compromise the BHA through corrosion or erosion, nor did they leave residual cement deposits inside the coiled tubing. As a result, the integrity of the technology was preserved, enabling the assembly to complete all three planned abandonment stages successfully, each repeated across the three zones under evaluation.

Conclusions and Way Forward

Compared to conventional rig-based abandonment operations, the rigless methodology reduced intervention time by several days, minimized mobilization costs, and significantly lowered surface footprint and HSE exposure, while still ensuring barrier integrity in full compliance with regulatory standards.

As expected in a pioneering project of this nature, several areas for improvement were identified that can further reduce execution time and enhance the economic and environmental sustainability of rigless interventions compared to conventional rig-based operations. Among these, the adoption of a custom-engineered 9" QTS in the pressure-control equipment configuration would enable faster BHA change-outs while accommodating diameters greater than 8", thereby streamlining multi-run programs.

From a technological standpoint, the full potential of the HCTT platform can be further leveraged by exploiting its fiber-optic capabilities for Distributed Temperature Sensing (DTS) and Distributed Acoustic Sensing (DAS). This would allow direct comparison with PLT results and, in future campaigns, potentially eliminate the need for a dedicated PLT run by acquiring injection profiles across the entire zone during other operational passes.

The success of this pilot campaign has led to the rigless methodology being confirmed for the remaining wells in the SCAN project. Building on the lessons learned, continuous improvement will remain a core driver, ensuring that future applications achieve even greater efficiency, lower carbon footprint, and enhanced data quality.

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